SMITH ENGINEERING

Queen's University

Department of Electrical and Computer Engineering



ELEC 490/498 Project Blueprint

Listen to Plants: A Multichannel Electrical Sensing Platform for Plant Physiology

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Executive Summary

Passive electrophysiology monitoring and measurement of leaf bioimpedance via electrical impedance spectroscopy (EIS) exist separately, however a low-cost, leaf-wearable system combining both capabilities does not. These metrics are important for various scientific ventures, both in industry and academia, thus a targetable market gap exists. Greenhouse operators seek to improve yield, and characterizing their crops greatly simplifies that process. Electrical plant-state data is a key input to creating diagnostics on plant-stress, water-status, and light-deprivation. As such, the goal is to produce reliable, synchronized, machine learning (ML)-ready electrical measurements and metadata. The end product will be a 4-electrode leaf clip, a switchable dual-path printed circuit board (PCB) (passive and EIS), and a Raspberry Pi (PI) logger. The data will be validated against gas exchange data collected by an LI-6400 in a 168 hour greenhouse deployment to ensure correctness and accuracy.

The proposed design comprises two measurement modes, which alternate periodically. The passive path will feature continuous recording with DC coupling, a sampling rate of at least 256 Hz, and a 1 MΩ differential input impedance. The EIS path will inject an alternating current (AC) current excitation through the four-electrode clip and measure impedance at five frequencies varying from 1 to 100 kHz every 15 $\pm$1 min. The two paths will be integrated using known hardware components, verified using known resistor–capacitor (RC) circuits, then integrated on PCB revision (Rev) A, while Rev B will reduce noise found during LI-6400 testing.

Development will produce five prototypes; the clip, the passive path, the EIS path and PCB revisions A and B. Testing will occur in stages, starting with individual component validation against controlled sources and completed paths validated against RC networks with known values. The integrated design will then be validated against an LI-6400 during a 7 day (168 hour) deployment in a greenhouse environment. Lab equipment, 3D printers, an LI-6400, and greenhouse space will be provided by the project supervisors, while necessary electronic components will be ordered as needed. The estimated total project cost is $457 CAD. The end result of the project will be a deployable prototype on a custom PCB that produces accurate data, ready for integration with greenhouse control systems, or training ML algorithms.

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Acronyms & Abbreviations

AC	alternating current
ADC	analog-to-digital converter
AFE	analog front end
BOM	bill of materials
CAD	computer-aided design
CEC	Canadian Electrical Code
DC	direct current
EIS	electrical impedance spectroscopy
FSR	force-sensing resistor
GFCI	Ground-Fault Circuit Interrupter
I^2C	Inter-Integrated Circuit
IP65	Ingress Protection 65
ML	machine learning
PCB	printed circuit board
PI	Raspberry Pi
QC	quality control
RC	resistor–capacitor
Rev	revision
SPI	Serial Peripheral Interface

1. Introduction

1.1 Problem and Purpose

This project aims to create a low-cost, ready-to-use, reliable plant monitoring system to be deployed in a greenhouse environment. Most current systems measure environmental conditions rather than physiological responses and pose significant cost barriers for practical applications [1] [2]. By developing an inexpensive, simple system that measures electrical signals from plant leaves, horticulturists, researchers and even hobbyists can monitor their plants' biological state more precisely. These measurements also allow researchers to cross-validate their analytical results with other systems, such as the LI-6400, leading to more reliable interpretations of plants' physiological responses.

1.2 Project Objective and Scope

The objective of the project is to design a compact and inexpensive sensor that is capable of remaining attached to a leaf for at least 72 hours without causing damage to the leaf [3]. The system's end goal is reliable timestamped output data that can be analyzed by scientists or horticulturists. To ensure reliability, an LI-6400 will be used to cross reference data. The LI-6400 measures photosynthesis, transpiration and stomatal conductance based on CO_2 and $\mathrm{H}_2\mathrm{O}$ gas concentrations [4]. The operating conditions are $18 - 30°$C.

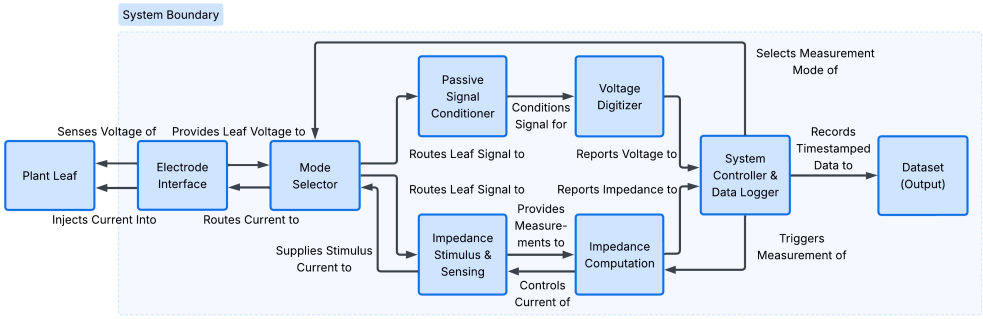


Figure 1: Logical Flow for the Plant Monitoring System

Figure 1 shows the logical flow from the leaf electrodes through the passive and EIS paths to the PI, where data is logged and synchronized. The project scope covers the electrical hardware and data-processing software for a low-cost plant monitoring system. The hardware provides two measurement paths, passive voltage and EIS. The software compiles the measured data into a time-stamped dataset with metadata. Using this data to train ML models for greenhouse control is outside the project scope.

1.3 System Deliverables

The deliverables are a lightweight 3D-printed leaf clip with built-in electrodes, a custom PCB implementing the measurement paths, and a PI running the control and data-logging software. The system must be deployable in a greenhouse, log data continuously, with results correlating to simultaneous LI-6400 measurement fluctuations [5].

2. Functional Description

2.1 General System Overview

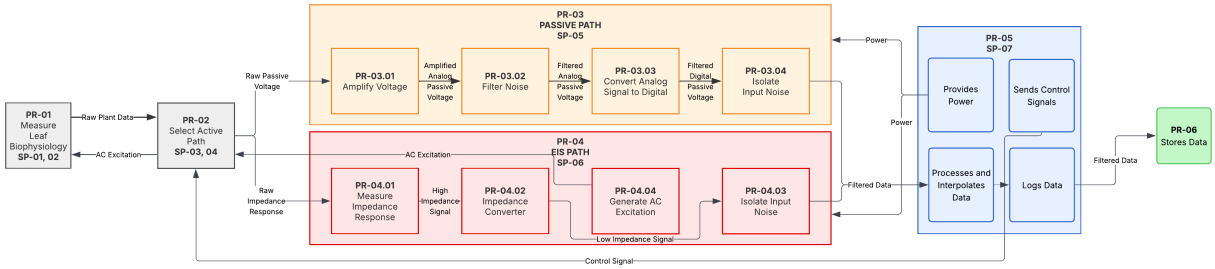


Figure 2: Functional decomposition diagram for the Plant Monitoring System

Figure 2 illustrates the functional decomposition of the system. PR-01 represents the leaf clip, which attaches to a leaf for taking measurements. The clip will contain four electrodes and a reference to enable correct measurement of passive electrical signals [1] and periodic EIS [2]. PR-02 represents the switch box that is driven by the PI. PR-03 represents the passive path and its sub-components which are responsible for measuring the potential difference between the probes, amplifying the signal, filtering noise, converting the analog signal to digital and isolating Serial Peripheral Interface (SPI) noise that the Raspberry Pi (PR-05) could be generating. PR-04 is responsible for the EIS path, generating excitation frequencies, measuring the amplitude and phase of impedance, converting the impedance measurement to usable data, and isolating Inter-Integrated Circuit (I^2C) noise that the PI might be generating. PR-06 is an SD card storing the timestamped data. An expanded view of Figure 2 can be found in Appendix A.

2.2 Requirement Specifications

Table 1 defines the key requirement specifications for the leaf monitoring system. Verification and validation, in Section 4, will ensure the system meets the requirements.

Table 1: Requirement Specifications

ID	Requirement	Weight
SP-01	Leaf clip shall remain attached for a period of 72 hours without causing visible necrosis/crushing (VT-02, VT-09)	Major
SP-02	During the attachment period the clip shall maintain continuous contact between all electrodes and the leaf (measurements should not have gaps other than when mode switches) (VT-02, VT-04, VT-07)	Critical
SP-03	Mode switching shall determine which path is active. The EIS Path shall be active every 15 minutes for 1 minute, switching within 15ms of the expected period (VT-06, VT-09)	Minor
SP-04	During the testing period a path shall always be active. A continuous data-flow with gaps up to 1ms is considered acceptable (VT-05, VT-06)	Major
SP-05	The passive path shall measure voltage readings from known RC circuits with 1 mV of acceptable noise (VT-01, VT-08)	Critical
SP-06	The active EIS path shall measure impedance magnitude and phase from known RC circuits within $\pm 5\,\%$ in magnitude and $\pm 2^{\circ}$ in phase at 1, 10, 30, 70, and 100 kHz (VT-03, VT-08, VT-09)	Critical
SP-07	The Raspberry Pi shall be continuously logging output data and timestamps to an SD card (VT-05, VT-08, VT-09)	Critical
SP-08	Using the data collected an AI model may be trained. The model may interpret the system's output data as plant physiological responses.	Stretch

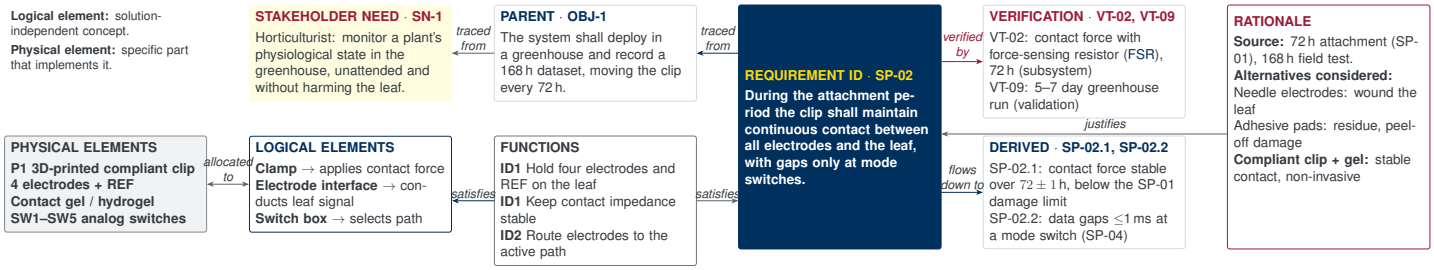


Figure 3: Critical requirement breakdown for SP-02.

Figure 3 shows a high level breakdown of the critical requirement for SP-02. A higher level breakdown for SP-02, as well as breakdowns for SP-05, SP-06 and SP-07 can be found in Appendix B.

2.3 Prototypes

Table 2 lists the prototype stages and their respective hardware configurations. Each prototype is also linked to the related success criteria, and milestones in Section 3.

Table 2: Prototype identification and characterization.

ID	Prototype	Hardware and configuration
P1	Leaf clip	3D-printed clip with 4 electrodes and a reference electrode, sized to fit the LI-6400; contact gel or hydrogel interface.
P2	Passive bench chain	Passive frontend amp with a low-pass and notch filter connected to the 24-bit analog-to-digital converter (ADC); bench supplied voltage. A PI logging script reads the ADC over SPI.
P3a	Bench EIS	Function generator and oscilloscope driving a four-electrode sweep from 100 Hz to 1 MHz on RC dummy networks.
P3b	AD5933 EIS	AD5933 EVAL board with a four-electrode front end, measuring 1–100 kHz on the same RC networks.
P4	PCB Rev A	Both measurement paths on one board with analog mode switching, isolated power, isolated SPI/I^2C, and a PI header.
P5	PCB Rev B	Rev A with the noise fixes from LI-6400 testing: shielding, single-point ground, and a battery-powered front end if required.
P6	Field system	P5, P1, and the PI logger in a greenhouse enclosure.

2.4 Risks and Safety Hazards

The primary risk is moisture entering the electrical system, causing short circuits, electric shock, or equipment failure. The greenhouse environment falls under section 22, category 1 of the Canadian Electrical Code (CEC) [6]. Electrical systems, including the PI, integrated PCB and filtering components, will be placed in an enclosure with a rating of Ingress Protection 65 (IP65), to prevent dust and moisture from damaging equipment or users [7]. Cables will be sealed, and Ground-Fault Circuit Interrupter (GFCI) protected supplies will be used to further protect the system [8]. The secondary risks arise from overheating of electrical systems from sunlight and restricted airflow. To mitigate this, the system will be placed away from sunlight and will contain a 5 A fuse in the main connection line to prevent over-current, as recommended by the Raspberry Pi Foundation [9]. Finally, the recommendations outlined in the University of Florida Greenhouse Safety Manual will be followed to ensure that all preventable damage and injuries are avoided [10].

3. Design and Production Approach

3.1 Development process

The first step will be to reproduce the measurement modes seen in the literature on evaluation hardware and verify it against dummy loads [1], [2]. The subsystems are then combined on the Rev A PCB. Rev B will be an iteration of Rev A correcting for any noise problems found during LI-6400 testing.

The passive path uses an INA121 instrumentation amplifier [11] and an AD7124-8 ADC board [12]. It is direct current (DC) coupled so that slow potential changes are not filtered out [13]. For EIS, the four-electrode measurement is first swept from 100 Hz to 1 MHz with the lab function generator and oscilloscope [2]. The same RC networks are then measured with the EVAL-AD5933 board [14] from 1 to 100 kHz. Table 3 lists the prototypes and the associated requirements, tests, and milestones.

Table 3: Prototype purpose and traceability.

ID	Purpose	Requirements	Tests	Milestones
P1	Check electrode contact and LI-6400 fit; reprint if necessary.	SP-01, 02	VT-02, 04, 07	M1, M8
P2	Rebuild a published passive recording setup as the reference for P4.	SP-03, 04	VT-01	M2
P3a	Measure the RC networks from 100 Hz to 1 MHz to characterize impedances.	SP-05	VT-03	M3
P3b	Check the AD5933's accuracy before it goes on the PCB.	SP-05	VT-03	M3
P4	First board with both paths, mode switching, and logging, compared against P2 and P3.	SP-03–07	VT-01, 03, 05–08	M4–M6
P5	Rev A with the noise fixes from M6.	SP-03, 05–07	VT-01, 3/8	M7
P6	Rev B and the clip running in the greenhouse for 72 h.	SP-02, 06, 07	VT-09	M8

3.2 Milestones and division of labor

Table 4 outlines the project milestones, prototypes, and due dates. Elizabeth is responsible for the clip and Rev A (M1, M4, M5) and builds Rev B. Trevor leads the passive path and the PI logger (M2), and Hubert leads the EIS path (M3). Foster leads the LI-6400 and greenhouse testing (M6–M8).

Table 4: Milestones, due dates (2026–27), and status as of 1 Nov 2026.

ID	Milestone and completion evidence	Prototype	Tests	Due	Status
M1	Clip fits the LI-6400; causes no leaf decay (24 h).	P1	VT-02, 04	30 Oct	Delayed, clip still in design.
M2	Passive chain works on a dummy source, then records a live plant.	P1, P2	VT-01	7 Nov	Dummy-source test done. Live plant waits M1.
M3	Bench sweep and EVAL-AD5933 within $\pm 5\,\%$ in $\lvert Z \rvert$ and $\pm 2^\circ$ in phase.	P3a, P3b	VT-03	14 Nov	Not started
M4	Rev A design ordered.	P4	—	7 Dec	Not started
M5	Rev A modes agree with P2 and P3 within tolerance; logging, mode switching, and wiring work.	P4	VT-01, 03, 05–07	22 Jan	Not started
M6	Passive noise floor changes by ≤ 3 dB in the LI-6400.	P4	VT-08	24 Jan	Not started
M7	Rev B passes bench checks; recorded data aligns with LI-6400 to ≤ 1 s.	P5	VT-01, 03, 08	19 Feb	Not started
M8	On leaf greenhouse deployment completed. (72 h)	P6	VT-09	12 Mar	Not started

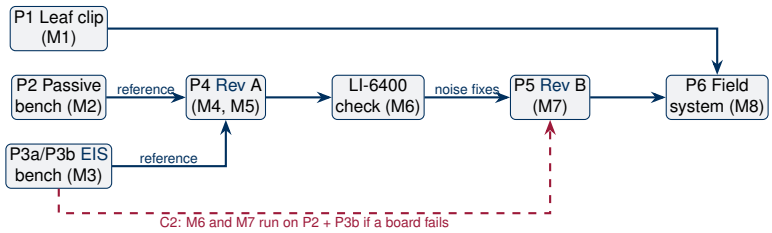


Figure 4: Dependencies between prototypes and milestones. The dashed path is contingency C2.

Figure 4 shows how the prototypes depend on each other. The clip and the two bench circuits (M1–M3) are built in parallel. Their results become the reference for Rev A, which is not accepted until both modes agree with them (M5). An initial LI-6400 noise check (M6) decides what is iterated for Rev B, and Rev B is used for the LI-6400 validation recording (M7) and the greenhouse run (M8).

3.3 Contingencies

Table 5 includes the risks the project faces and the milestones they affect. The main risk is a manufacturing delay of PCB Rev A or B causing a delay in the LI-6400 testing.

Table 5: Contingency plan.

ID	Risk and trigger	Response	Affected	Impact
C1	LI-6400 or mains noise [5]. Passive noise floor rises by more than 3 dB with the LI-6400.	Add shielding, a single-point ground, and a battery-powered front end on Rev B.	M7	+1–2 wk, taken from M8 float; up to $30 from headroom.
C2	Rev A fails the M5 tolerances, or Rev B is unassembled by 6 Feb.	Run M6–M7 on P2/P3b with manual mode switching during debugging.	M5–M7	LI-6400 testing on schedule; $0
C3	Electrical safety. Condensation forms inside the enclosure, or mains equipment is within reach of irrigation.	Run the front end and PI on battery only, or use a GFCI-protected outlet.	M8	$0–$40 from headroom.
C4	LI-6400 or greenhouse booking is not confirmed 2 wk before M6, M7, or M8.	Book when M5's date is known. If a booking slips, cut the M7 recording to one light-response curve.	M6–M8	Less data; $0

4. Testing and Evaluation

The testing is divided into five stages, illustrated in Figure 5 and expanded in Appendix C. Verification occurs primarily on a test bench through component verification, testing the operating characteristics of the components; subsystem testing (blue), guaranteeing performance metric satisfaction; and integration testing (green). These tests confirm that both system-level requirements and individual functionality meet expectations. Validation will primarily occur through LI-6400 testing (red) and greenhouse testing (purple), validating prolonged operation and plant vulnerability. Validation alongside the LI-6400 will show that changes in environmental factors, such as irrigation, can produce readings comparable to the 85 mV change reported in Gil et al. and correlated with independently measured physiological responses, such as stomatal conductance [15]. Establishing these relationships would validate that this system captures relevant passive and EIS readings. Table 6 further defines these tests and their outcome metrics.

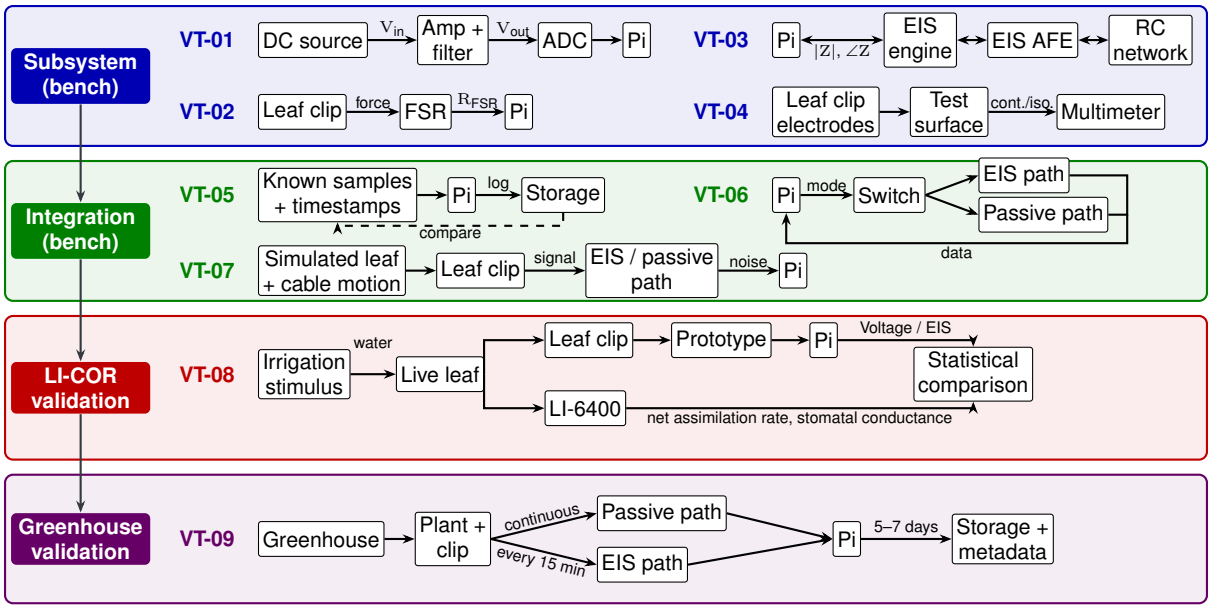


Figure 5: Staged testing from subsystem to greenhouse verification and validation.

Table 6: Proposed system verification and validation tests.

Requirement	Proposed test	Expected evidence of success
VT-01| P2, P4, & P5 | SP-03 & SP-04	Apply a known DC input to the passive AFE; compare with ADC output.	$\geq$256 Hz continuous sampling, DC/Near DC response, 1 ± 0.01 MΩ differential input impedance.
VT-02 | P1 | SP-02	Measure the clip's contact force using an FSR sensor.	Reproducible, stable contact for 72 ± 1h without leaf damage.
VT-03 | P3a, P3b, P4, & P5 | SP-05	Measure a known RC circuit within the EIS subsystem at select frequencies.	Impedance, magnitude, and phase values at 1, 10, 30, 70, and 100 kHz with $\pm$ 5 % impedance magnitude and $\pm$ 2deg phase tol.
VT-04 | P1 | SP-01	Apply a conductive load to the electrodes and measure the output with a multimeter.	Readings match the known conductive load with no measurable continuity (open-circuit) between designated isolated connections.
VT-05 | P4 & P5 | SP-07 & SP-03	Feed a known dataset to the Pi Logger and verify logged data, metadata, and QC information.	$\geq$256 Hz sampling with correct timestamps, metadata and dropped value flags.
VT-06 | P4 & P5 | SP-06	Repeatedly switch EIS and Passive. Record the transition between modes.	Switching every 15 ± 1 min that doesn't trigger passive system events. Path isolation functions.
VT-07 | P1 & P4 | SP-01	Connect the clip to a conductive surface and read values the wires are fixed and moving.	Noise floor increases by $\leq$ 3dB versus the stationary wire.

VT-08 | P4 & P5 | SP-03, SP-05, SP-06, & SP-07	Operate the prototype alongside LI-6400. Record their baselines then irrigate the plant soil. Compare synchronized passive and EIS measurements against CO2 assimilation A and stomatal conductance $\mathrm{g_s}$ for a 40 min period. Repeat trial.	Noise floor increases by $\leq$ 3 dB. Irrigation produces an increase in A and $\mathrm{g_s}$[5]. The prototype detects repeatable electrical signal changes above the baseline, with $\sim$ 85 mV as reference [15].
VT-09 | P6 | SP-02, SP-06, & SP-07	Operate system within greenhouse environment for 5–7 days.	Continuous sampling at $\geq$ 256 Hz with EIS every 15 ± 1 min; clip attached to one leaf for 72 ± 1 hrs without damage or signal loss.

5. Resource Requirements

5.1 Scheduling access to resources

As seen in Table 7, 3D printing and PCB manufacturing have 1 day and 7 day lead times respectively, and access to the greenhouse and biology lab require 2 weeks of notice to schedule access. These constraints are factored into the deadlines of each milestone and a detailed schedule can be seen in Appendix D.

Table 7: Resource access by milestone.

Milestone	Resource / source	Access date / lead time
M1	3D Printing	Upon Request; 24 hr lead time
M2, M3	Sci 65' E.D. Lab	Accessible by Fob
M4, M7	PCB Manufacturing [16]	7 Day Lead Time including Shipping
M1, M6-M8	Greenhouse/Biology Lab Access	Accessible with 2-Weeks Notice

5.2 Materials and other resources

Table 8 shows a simplified budget including the provided lab resources, parts that have been purchased from DigiKey [17], and projected PCB manufacturing costs[16]. A more detailed bill of materials (BOM), including analog and digital component lists, can be found in Appendix E. The budget includes $100 for replacement or forgotten components as well as $143 of headroom below the allowed $600.

Table 8: Preliminary cost estimate (CAD).

Item / source	Qty.	Unit cost	Total
Electrode Clip 3D printing, Electrical Leads, Conductive Gels	1	Provided By Lab	$0
Digital & Analog Assembly / DigiKey[17]	1	$293.59 (HST)	$293.59
PCB Manufacturing/ PCBWay[16]	2	$31.55	$63.10
Reserve & Forgotten Components	1	$100	$100
Estimated total			$456.69

6. Conclusion

To address the cost and lack of physiological data, this project proposes a low-cost, leaf-wearable sensor combining passive plant electrophysiology and plant bio-impedance. The functional prototype integrates two measurement streams with a PI and data logger to collect synchronized, timestamped, and ML-ready electrical data and metadata. In combining both modes affordably, the system could make plant monitoring more accessible to hobbyists and growers. The 8 milestones range from subsystem development and PCB production to system validation with the LI-6400 and greenhouse environments. Each critical requirement links to a test and milestone SP-05 $\rightarrow$ VT-01/VT-08 $\rightarrow$ M2/M6. Early verification testing will ensure subsystem and integration requirements. Later validation assesses whether stakeholder needs are met, including synchronized data collection with the LI-6400. The project succeeds when Rev B and the clip complete the VT-09 greenhouse deployment and the M7 recording shows electrical signals tracking LI-6400 changes in A and $\mathrm{g_s}$. Feasibility is supported by the contingency budget, schedule float, university facilities, and the team's hardware experience. Currently, the leaf-clip is designed but delayed, and the passive path has been constructed on a breadboard. Next steps include clip completion and subsystem verification against system requirements. Several technical limitations could arise as electrode contact instability may compromise accurate and repeatable signal collection (SP-02, SP-05), and limited greenhouse testing may leave long-term failures undetected. Conductive gels and corrosion protection can help mitigate contact instability and degradation; however, prolonged reliability testing will require future work. Mitigating these risks will improve reliability and strengthen its contribution to plant monitoring.

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A. Detailed System Breakdown

Figure 6 is the expanded view of Figure 2 seen in Section 2. This illustration includes the parts used, their part numbers, and the signal flow between components.

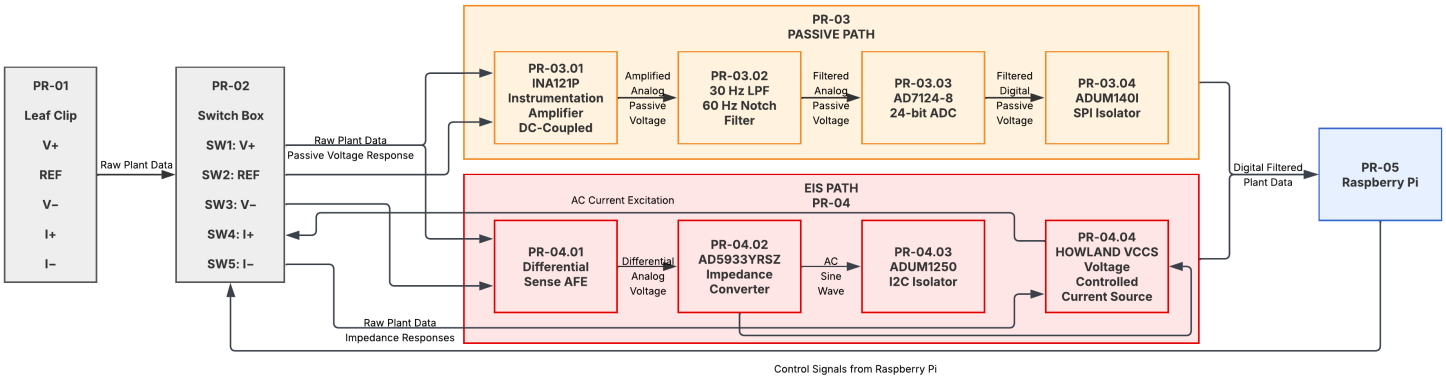


Figure 6: Expanded view of the system breakdown.

B. Critical Requirement Diagrams

Figures 7, 8, 9, and 10 trace each critical requirement from Table 1 to the stakeholder need and system objective they come from, the functions and logical elements that satisfy it, the physical parts it is allocated to, the tests that verify it (Table 6), and the sub-requirements it flows down to. Function numbers follow the IDs in Figure 2.

B.1 SP-02: Continuous Electrode Contact

Every measurement depends on the electrode–leaf interface. If an electrode lifts, the passive path records a step or open-circuit drift and the EIS path reports a contact impedance instead of the leaf's. Contact must also be gentle enough to meet SP-01, so the clip is a trade-off between contact force and leaf damage.

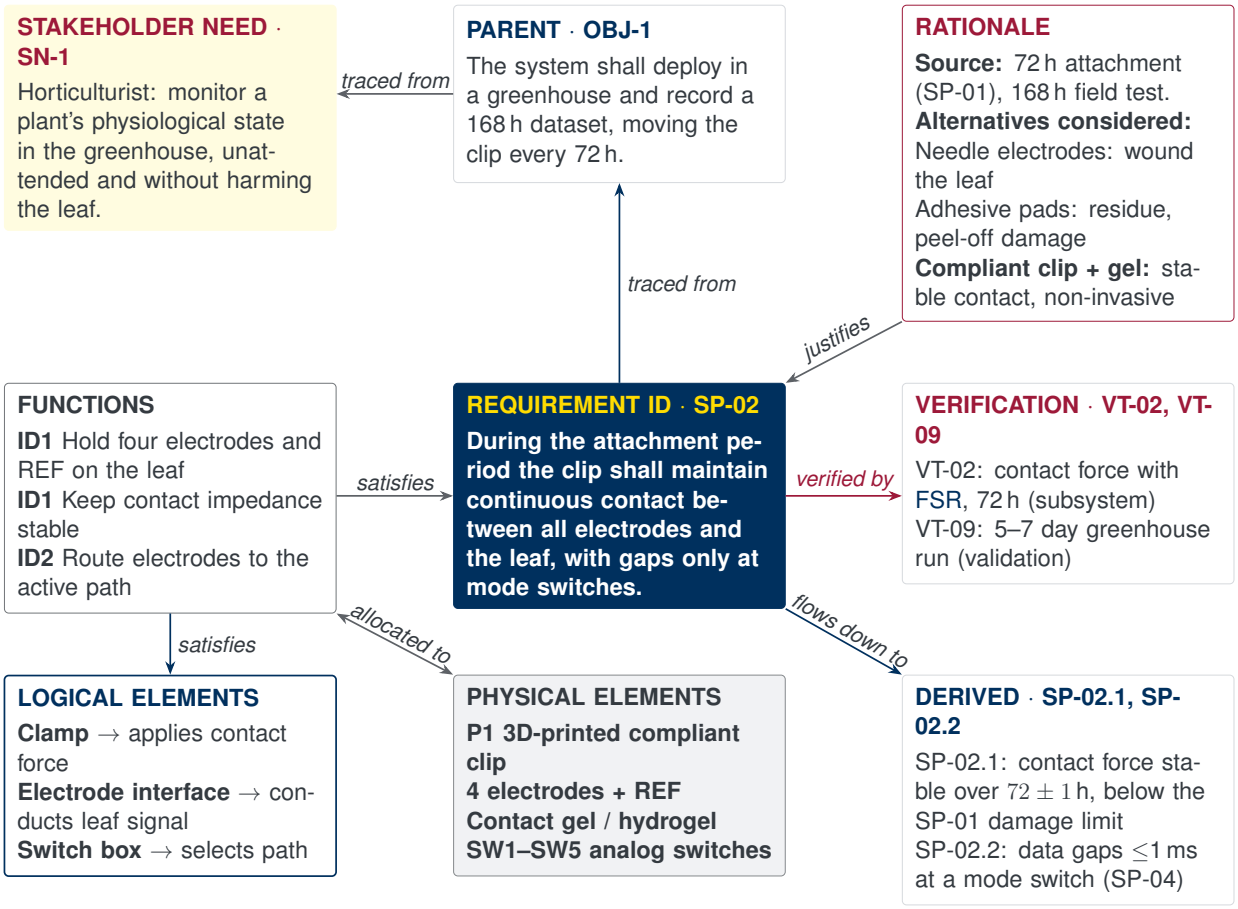


Figure 7: Critical requirement SP-02: continuous electrode contact.

B.2 SP-05: Passive Path Noise

Passive leaf potentials are slow and in the millivolt range [1], [13], so the noise floor of the passive path sets the smallest response the system can detect. The path is DC coupled so slow changes are kept, which also means it must reject mains pickup and digital noise from the PI.

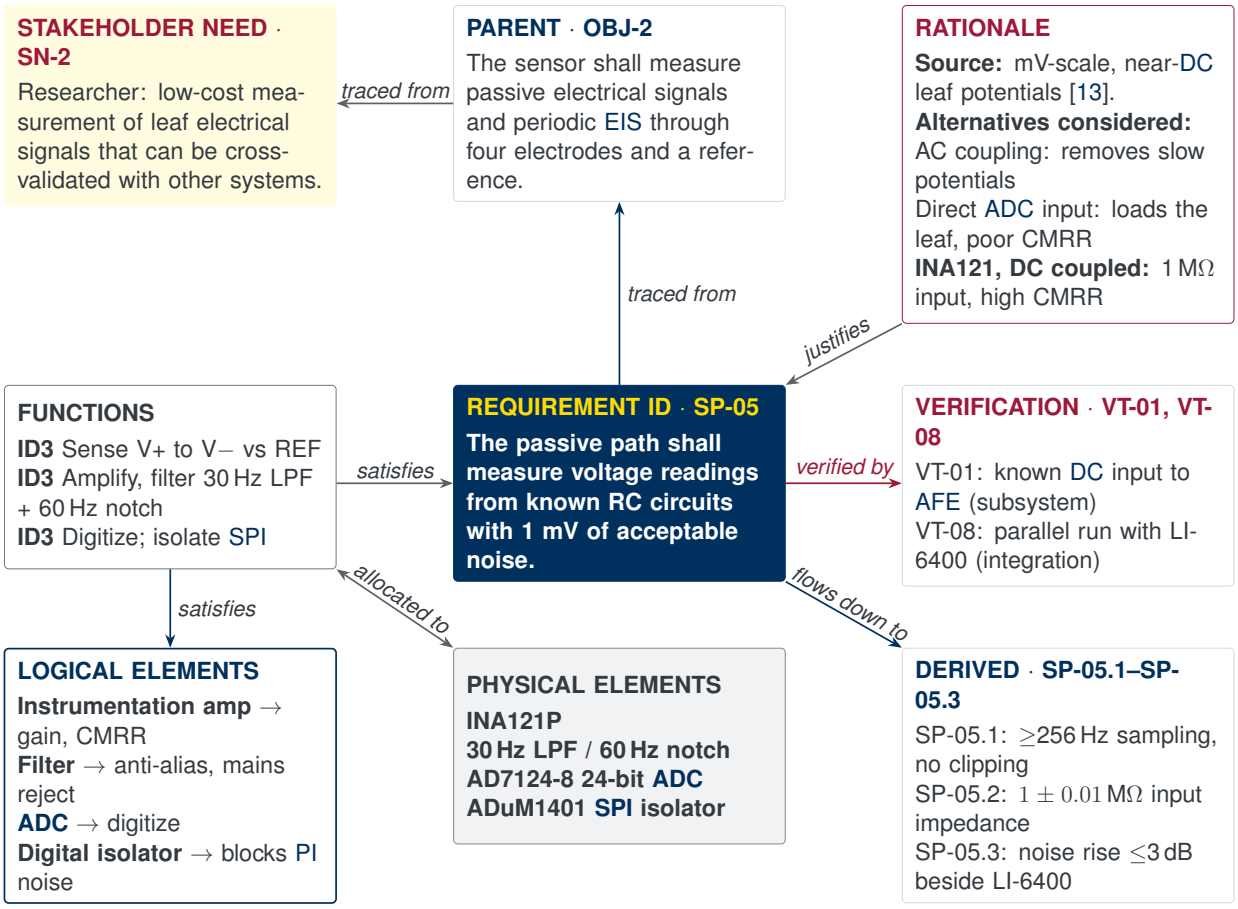


Figure 8: Critical requirement SP-05: passive path noise.

B.3 SP-06: EIS Accuracy

Leaf impedance magnitude and phase track tissue water and membrane state [2]. A four-electrode measurement separates the excitation and sense electrodes, so electrode polarization does not appear in the result. Accuracy is checked on known RC networks first, because their impedance can be calculated exactly.

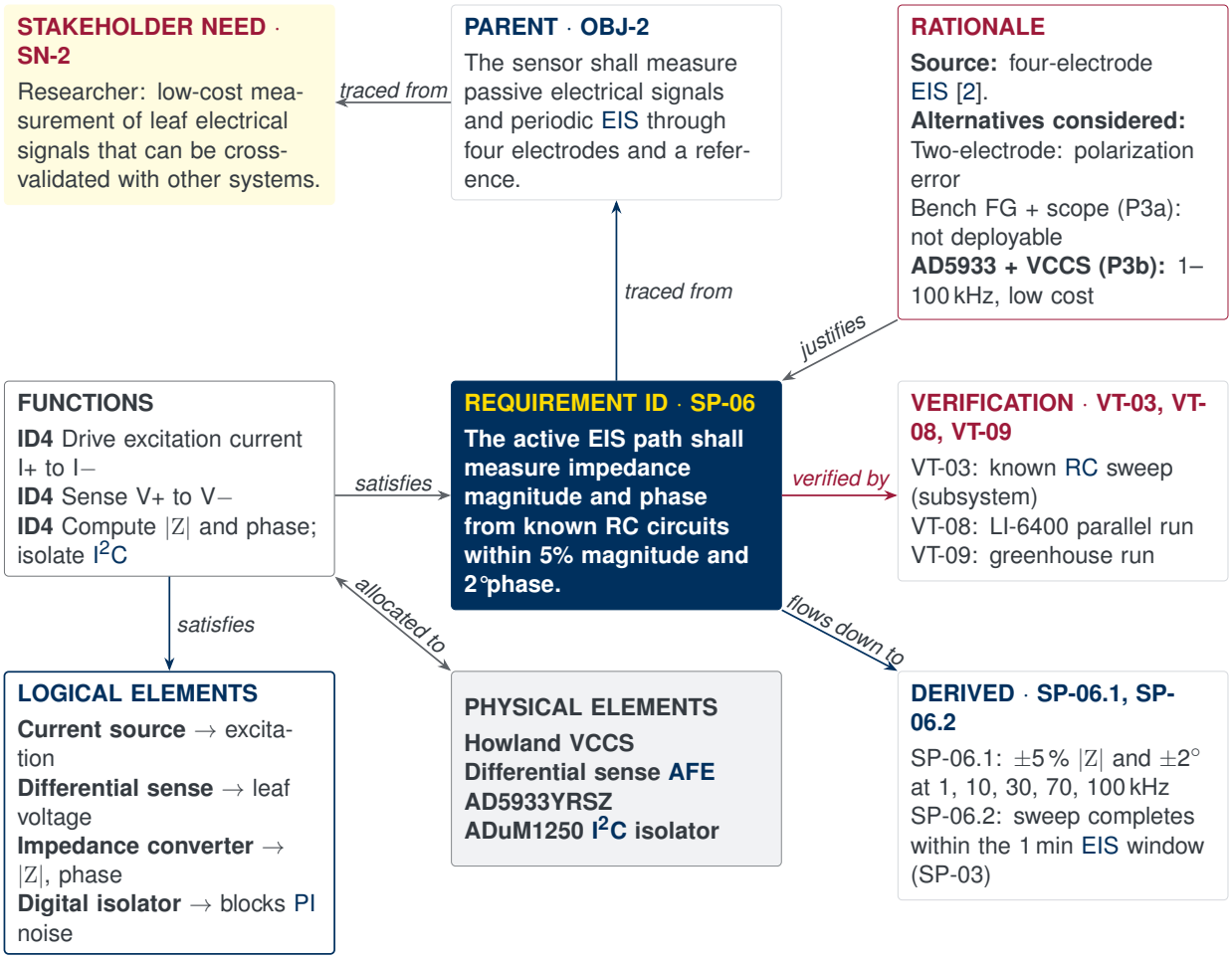


Figure 9: Critical requirement SP-06: EIS accuracy.

B.4 SP-07: Continuous Data Logging

The system's output is the dataset itself. Data that is not logged with a correct timestamp cannot be aligned with the LI-6400 or analyzed later, so logging has to run unattended for the full deployment and flag any samples it drops.

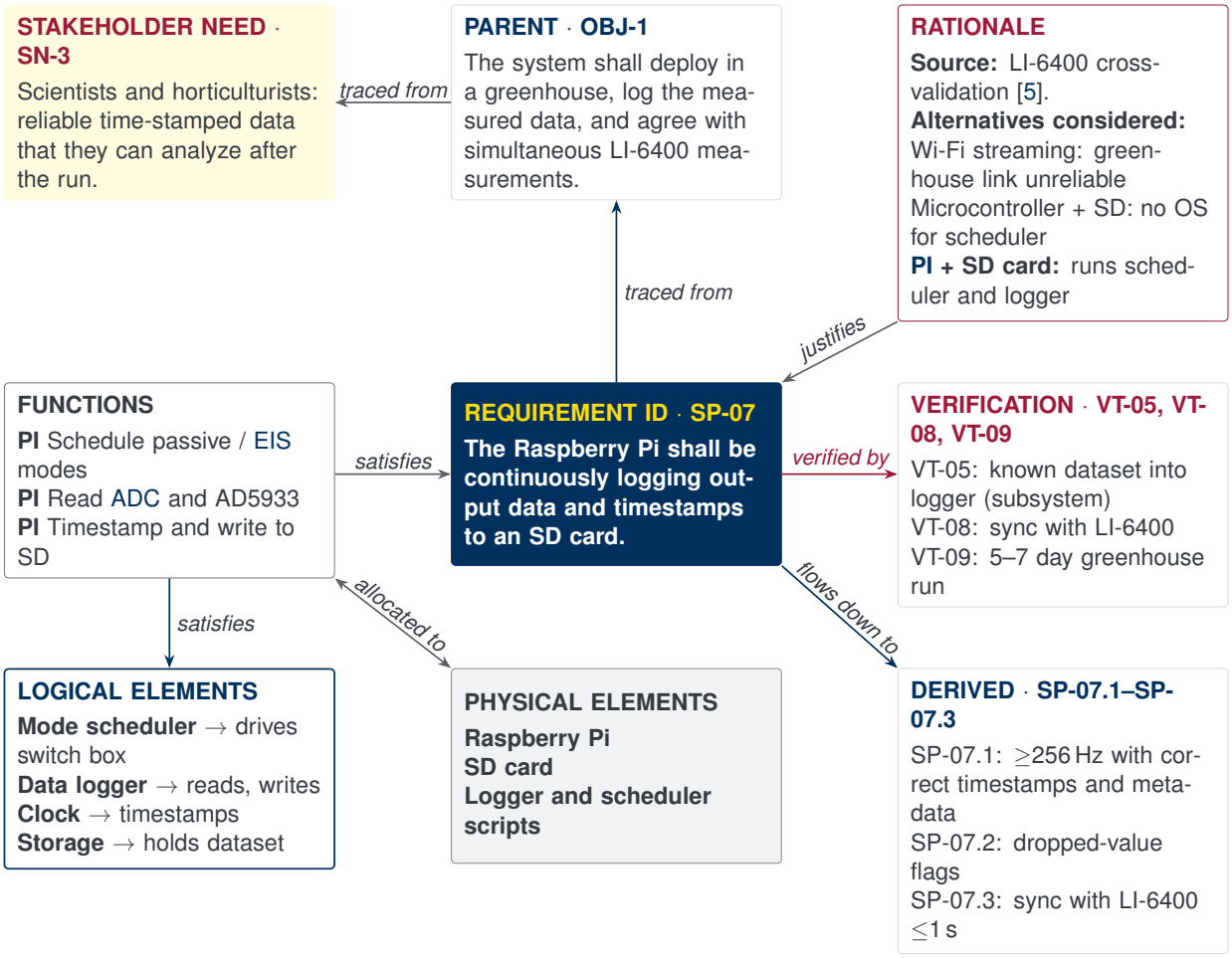


Figure 10: Critical requirement SP-07: continuous data logging.

C. Expanded Testing and Evaluation Table

Figure 11 is the expanded view of Figure 5 seen in Section 4.

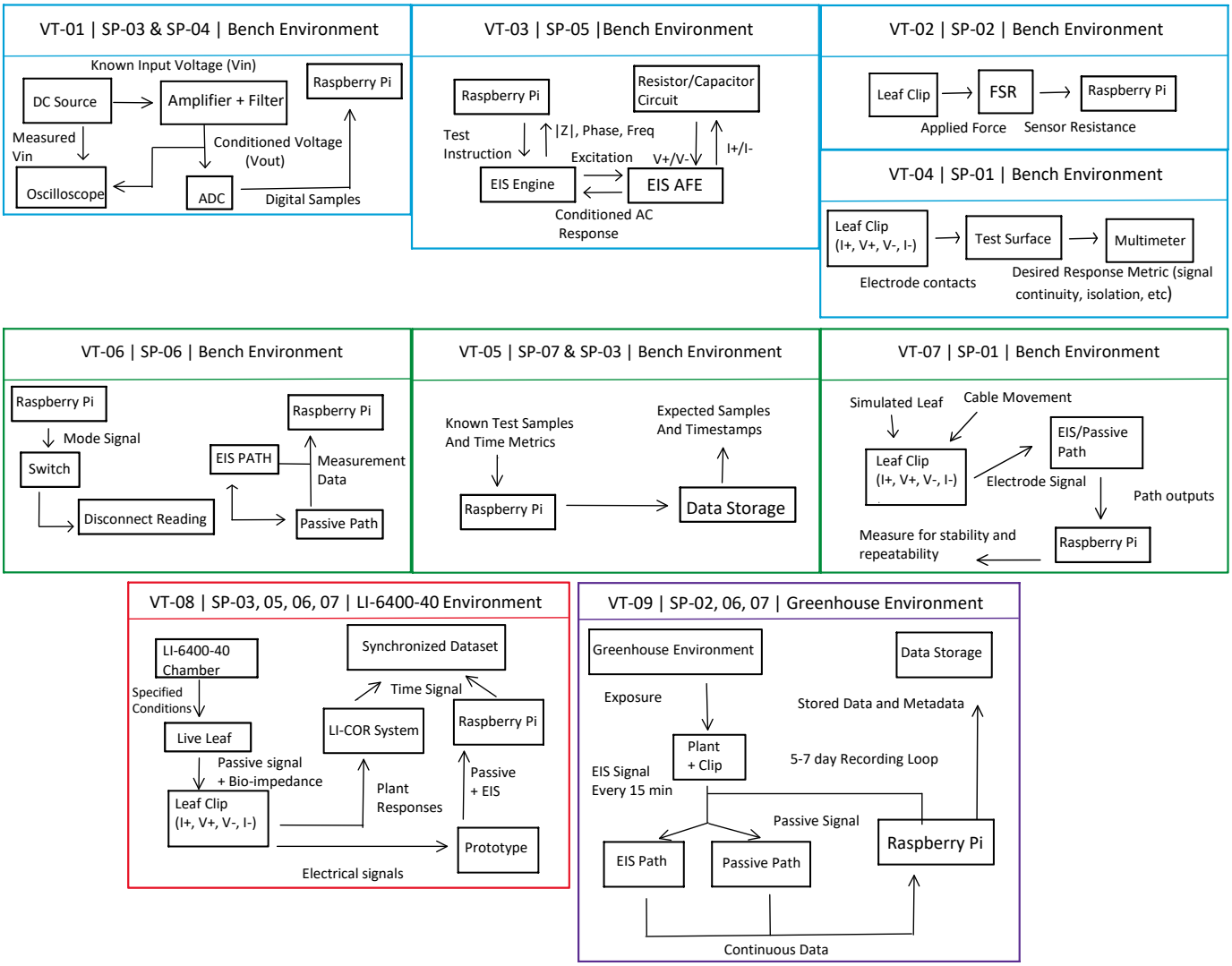


Figure 11: Expanded staged testing from subsystem to greenhouse verification and validation.

D. Detailed Project Schedule

Table 9 lists the planned start and finish date of each milestone, the due date, and the float. Float is the number of days between the planned finish and the due date. The schedule assumes work began on 1 October 2026 and continues seven days per week, with each milestone starting after the planned finish of the milestones it depends on.

Table 9: Project schedule with planned dates and float (2026–27).

Milestone	Owner	Start	Planned finish	Due	Float	Description
Subsystem Development (M1–M3)						
M1: Leaf clip	Elizabeth	1 Oct	23 Oct	30 Oct	7 d	Leaf clip is designed, printed and checked to fit the LI-6400 and causes no leaf damage after 24 h.
M2: Passive chain	Trevor	1 Oct	31 Oct	7 Nov	7 d	Passive chain is assembled and tested on a breadboard using EVAL board. Passive chain works on a dummy source, then records a live plant.
M3: EIS bench	Hubert	1 Oct	5 Nov	14 Nov	9 d	EIS path is assembled and tested on breadboard using EVAL board. Bench sweep and EVAL-AD5933 within $\pm 5\,\%$ in $\lvert Z \rvert$ and $\pm 2^{\circ}$ in phase of the calculated RC values.
PCB Rev A (M4–M5)						
M4: Rev A design and order	Elizabeth	6 Nov	1 Dec	7 Dec	6 d	Initial PCB design(Rev A) is created using computer-aided design (CAD) software. Rev A design reviewed and ordered from PCBWay.
M5: Rev A verification	Elizabeth	2 Dec	10 Jan	22 Jan	12 d	Rev A PCB arrives and is assembled using a mix of board-components and through hole components. Rev A modes agree with P2 and P3 within tolerance; logging, mode switching, and wiring work.

Milestone	Owner	Start	Planned finish	Due	Float	Description
Validation and Deployment (M6–M8)						
M6: LI-6400 noise check	Foster	11 Jan	17 Jan	24 Jan	7 d	Rev A PCB is tested in the biology lab with the LI-6400. Passive noise floor changes by $\leq$3 dB with the LI-6400 on vs. off.
M7: Rev B and LI-6400 recording	Elizabeth / Foster	18 Jan	12 Feb	19 Feb	7 d	Rev B PCB is designed as an iteration of Rev A, to improve noise performance. Rev B is ordered, delivered, and assembled. Rev B passes the bench checks, then records alongside the LI-6400 with both datasets aligned to $\leq$1 s and no interference.
M8: Greenhouse deployment	Foster	13 Feb	26 Feb	12 Mar	14 d	Rev B PCB is assembled with the final leaf clip prototype and PI set to record for a long-term greenhouse test. 5–7 day greenhouse deployment completed.

E. Detailed Bill of Materials

Table 10 lists the full bill of materials, grouped into lab-provided equipment, prototype hardware, and production and reserve allowances. Unit costs are shown before tax in CAD unless a different currency is stated. Items purchased by students list their supplier, and the Status column gives the procurement status of each line. All prices are before tax (HST) in CAD.

Table 10: Bill of materials (CAD).

Item / candidate	Qty.	Unit cost	Supplier	Status	Reason
Lab & Validation Components					
LI-6400XT / fluorescence head / greenhouse access	1	Already available	Provided by lab	Accessible in the Lab	Physiological validation and controlled light/CO_2 experiments.
Oscilloscope, function generator, DMM, bench supplies	1	Existing EE lab equipment	Provided by lab	Accessible in the Lab	Lets the team reproduce Serrano-style EIS and characterize noise without buying a USB instrument.
3D printer + filament	1	Existing lab/course facility	Provided by lab	Accessible in the Lab	Simple compliant leaf clip and enclosure prototypes.
Optional Plant SpikerBox	1	$209.93	Optional lab benchmark	Not Received	Useful comparator/ open-schematic educational instrument; not required for the final prototype.
Prototype Hardware					
Raspberry Pi + existing communication interface	1	Already available	Provided by Vince L, ECE Bain Lab	Received	Wired logging/time synchronization; wireless is not required for the capstone.
Passive front end: INA121P instrumentation amplifier	1	$19.29	Student Purchase (DigiKey [17])	Received	Vivent-style high-Z differential acquisition
Passive ADC: EVAL-AD7124-8-PMDZ	1	$74.17	Student Purchase (DigiKey [17])	Received	24-bit ADC development board; bare IC can replace later

Item / candidate	Qty.	Unit cost	Supplier	Status	Reason
Passive ADC: AD7124-8BCPZ	1	$25.99	Student Purchase (DigiKey [17])	Not Ordered	24-bit ADC bare IC; funded from headroom if used.
EIS engine: EVAL-AD5933EBZ	1	$122.79	Provided by lab	Not Received	Fast EIS proof-of-concept and firmware reference; final prototype uses bare AD5933.
EIS engine: AD5933YRSZ	1	$42.86	Student Purchase (DigiKey [17])	Received	Low-cost integrated impedance converter
Mode switching: ADG1219 low-leakage SPDT	5	$9.28	Student Purchase (DigiKey [17])	Received	Switch electrodes between passive/EIS modes
EIS / buffer AFE: OPA2140 dual JFET op-amp	3	$13.99	Student Purchase (DigiKey [17])	Received	0.5 pA input-bias class; useful for high-Z buffering
Isolated analog power: RH-0505D isolated ± 5 V DC/DC	1	$15.04	Student Purchase (DigiKey [17])	Received	Helps manage grounding/noise
Digital isolation: ISO1540DR I^2C isolator	1	$12.61	Student Purchase (DigiKey [17])	Received	Electrical separation to host where needed
Digital isolation: ADuM1401 SPI isolator	1	$7.47	Student Purchase (DigiKey [17])	Received	Electrical separation to host where needed
Fine bare silver wire, WPI	1	$67	Lab consumable	Not Received	Enough material for many Ag/AgCl-compatible leaf contacts; used across multiple groups/experiments.
SignaGel 60 g	1	$12	Lab consumable	Not Received	Conductive contact medium; one tube supports many trials.
Production Costs & Reserve Budget					

Item / candidate	Qty.	Unit cost	Supplier	Status	Reason
PCB fabrication: Two PCB iterations	2	$31.55	Allowance (PCB-Way [16])	Un-Used Allowance	Allows one learning/fix cycle
Contingency: Replacement/ forgotten parts	1	$100.00	Reserve	Un-Used Allowance	Covers passives, cables/connectors, enclosure/clip hardware, and replacement parts